

Varatharaju Mithuna

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EDUCATION

National University of Singapore

Bachelor of Computing in Computer Science, Minor in Interactive Media Development

Aug. 2023 – May 2027

EXPERIENCE

Frontend Engineering Intern

Jul. 2026 – Present

Power Instruments Pte Ltd

Singapore

- Developed and enhanced frontend features for the *EgenKit* vessel fuel monitoring platform using React, TypeScript, Redux, and Tailwind CSS.
- Built notification management features from UI to API integration, creating reusable components and implementing unit and end-to-end tests with Vitest, React Testing Library, and Playwright.
- Integrated backend APIs into the report settings feature, extending existing components and developing new reusable UI components with comprehensive test coverage.
- Migrated an existing feature to the new platform architecture, redesigning the UI and restructuring the implementation by replacing Redux state management with Zustand and introducing Zod schema validation.
- Collaborated with engineers in an Agile environment using Git-based workflows, participating in code reviews and incorporating pull request feedback.

Software Engineering Intern

May 2025 – Aug. 2025

Integro Technologies Pte Ltd

Singapore

- Developed SQL scripts to combine data from multiple database tables and extract data into importable scripts for internal applications.
- Created stored procedures in Oracle and PostgreSQL to support data import and processing workflows.
- Tested and debugged SQL scripts and stored procedures to ensure correct data extraction and processing.

PROJECTS

PeerPrep | *React, Node.js, Express.js, PostgreSQL, Prisma*

- Developed the Question Service and Question Attempt History Service for a collaborative coding interview platform using React, Express.js, PostgreSQL, and Prisma ORM.
- Designed the question database schema and built RESTful APIs for the frontend to retrieve and manage coding questions.
- Implemented real-time question attempt tracking, persisting users' coding attempts as they progressed and enabling them to resume and view their previous attempts.
- Built the frontend interface for viewing question attempt history and integrated it with the backend services.

Personal Portfolio | *React, TypeScript, Tailwind CSS, Vite*

- Designed and developed a responsive personal portfolio website to showcase software engineering projects, technical skills, and artwork.
- Built reusable React components and interactive UI elements using TypeScript and Tailwind CSS, and integrated a Gemini API-powered chatbot to explore AI application development.
- Deployed the application online and integrated project links and GitHub repositories for easy access by recruiters.

E-Commerce Platform Testing | *Jest, React Testing Library, Playwright, MongoDB*

- Designed and implemented the testing module for a full-stack e-commerce application, covering unit, integration, and end-to-end testing.
- Developed backend controller tests using Jest and MongoMemoryServer, and frontend tests using React Testing Library and Playwright to validate critical user workflows.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, Python, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, Express.js, Redux, Tailwind CSS, Prisma

Databases: PostgreSQL, Oracle, MongoDB, Supabase

Developer Tools: Git, Playwright, Jest, React Testing Library, IntelliJ IDEA, VS Code